LOVELY PROFESSIONAL UNIVERSITY

Lab Evaluation – III

DATA SCIENCE TOOLBOX – LABORATORY (CAP788)

Exploratory Data Analysis (EDA)

Submitted to: **Dr. Punam Rattan**
Submitted by: Rohit Kotamsetti | 12205208

Dataset: **Nike vs Adidas data from [Kaggle]**

Perform Exploratory Analysis on Student performance dataset and apply the following:

1. **Basic exploration of dataset:**

The initial phase of our analysis involved exploring the dataset to gain a better understanding of its contents. The dataset, named 'data.csv,' consists of information related to products from two major brands, Adidas and Nike. The dataset includes the following columns: Product Name, Product ID, Listing Price, Sale Price, Discount, Brand, Rating, Reviews, Gender, and Unique_Brand (a derived column indicating whether the product is for Men, Women, or Unisex).

[163]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv('data.csv')
```

[164]:
```
print(df.head())
```

```
                                        Product Name Product ID  \
0  Women's adidas Originals NMD_Racer Primeknit S…     AH2430
1                Women's adidas Originals Sleek Shoes     G27341
2                   Women's adidas Swim Puka Slippers     CM0081
3   Women's adidas Sport Inspired Questar Ride Shoes     B44832
4            Women's adidas Originals Taekwondo Shoes     D98205

   Listing Price  Sale Price  Discount                    Brand  Rating  \
0          14999        7499        50  Adidas Adidas ORIGINALS     4.8
1           7599        3799        50          Adidas ORIGINALS     3.3
2            999         599        40         Adidas CORE / NEO     2.6
3           6999        3499        50         Adidas CORE / NEO     4.1
4           7999        3999        50          Adidas ORIGINALS     3.5

   Reviews
0     41.0
1     24.0
2     37.0
3     35.0
4     72.0
```

```
4      72.0
```

[165]:
```python
print("Number of rows and columns:", df.shape)
```

```
Number of rows and columns: (3268, 8)
```

[166]:
```python
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3268 entries, 0 to 3267
Data columns (total 8 columns):
 #   Column         Non-Null Count  Dtype
```



```
---  ------         --------------  -----
 0   Product Name   3268 non-null   object
 1   Product ID     3268 non-null   object
 2   Listing Price  3268 non-null   int64
 3   Sale Price     3268 non-null   int64
 4   Discount       3268 non-null   int64
 5   Brand          3268 non-null   object
 6   Rating         2966 non-null   float64
 7   Reviews        3011 non-null   float64
dtypes: float64(2), int64(3), object(3)
memory usage: 204.4+ KB
None
```

None

[167]:
```python
print(df.describe())
```

```
       Listing Price    Sale Price     Discount       Rating      Reviews
count    3268.000000   3268.000000  3268.000000  2966.000000  3011.000000
mean     6868.020196   6134.265606    26.875765     3.572218    44.012953
std      4724.659386   4293.247581    22.633487     1.034382    30.455809
min         0.000000    449.000000     0.000000     1.000000     1.000000
25%      4299.000000   2999.000000     0.000000     2.800000    16.000000
50%      5999.000000   4799.000000    40.000000     3.700000    43.000000
75%      8999.000000   7995.000000    50.000000     4.400000    70.000000
max     29999.000000  36500.000000    60.000000     5.000000   223.000000
```

```
dtype: object
```

[171]:
```
df['Gender'] = df['Product Name'].apply(lambda x: 'M' if x.startswith(('Men',␣
↪'MEN')) else 'F' if x.startswith(('Women', 'WOMEN')) else 'Unisex')
print(df['Gender'].head())
```

```
0    F
1    F
2    F
3    F
4    F
Name: Gender, dtype: object
```

[172]:
```
gender_counts = df['Gender'].value_counts()
print(gender_counts)
```

```
Gender
M         1691
Unisex     803
F          774
Name: count, dtype: int64
```

[173]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3268 entries, 0 to 3267
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Product Name   3268 non-null   object
 1   Product ID     3268 non-null   object
 2   Listing Price  3268 non-null   int64
 3   Sale Price     3268 non-null   int64
 4   Discount       3268 non-null   int64
 5   Brand          3268 non-null   object
 6   Rating         3268 non-null   float64
 7   Reviews        3268 non-null   float64
 8   Gender         3268 non-null   object
dtypes: float64(2), int64(3), object(4)
memory usage: 229.9+ KB
```

[174]:
```
df['Unique_Brand'] = df['Brand'].apply(lambda x: 'Adidas' if x.startswith('A')␣
↪else 'Nike' if x.startswith('N') else 'Other')
print(df['Unique_Brand'].head())
```

3

```
0    Adidas
1    Adidas
2    Adidas
3    Adidas
4    Adidas
Name: Unique_Brand, dtype: object
```

[175]:
```
Unique_Brand_counts = df['Unique_Brand'].value_counts()
print(Unique_Brand_counts)
```

```
Unique_Brand
Adidas    2625
Nike       643
Name: count, dtype: int64
```

[176]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3268 entries, 0 to 3267
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Product Name   3268 non-null   object
 1   Product ID     3268 non-null   object
 2   Listing Price  3268 non-null   int64
 3   Sale Price     3268 non-null   int64
 4   Discount       3268 non-null   int64
 5   Brand          3268 non-null   object
 6   Rating         3268 non-null   float64
 7   Reviews        3268 non-null   float64
 8   Gender         3268 non-null   object
 9   Unique_Brand   3268 non-null   object
```

2. Handle missing values:

To ensure data quality and completeness, we addressed missing values in the 'Rating' and 'Reviews' columns. Null values in these columns were replaced with zeros, making the dataset ready for further analysis.

```
dtype: int64
```

[169]:
```
df['Rating'].fillna(0, inplace=True)
df['Reviews'].fillna(0, inplace=True)
```

[170]:
```
null_values = df.isnull().sum()
print(null_values)
```

```
Product Name     0
Product ID       0
Listing Price    0
Sale Price       0
Discount         0
```

2

```
Brand            0
Rating           0
Reviews          0
dtype: int64
```

3. Apply Normalization:

Normalization is a crucial data preprocessing step that helps in scaling numerical data to a common range. However, the report does not explicitly mention the specific normalization techniques applied to the data. It would be useful to know which columns were normalized and how they were transformed to a common scale.

```
[179]: from sklearn.preprocessing import MinMaxScaler
numerical_columns = ['Listing Price', 'Sale Price', 'Discount', 'Rating',␣
 ↪'Reviews']
scaler = MinMaxScaler()
df[numerical_columns] = scaler.fit_transform(df[numerical_columns])
print(df.head())
```

```
                                      Product Name Product ID  \
0  Women's adidas Originals NMD_Racer Primeknit S…     AH2430
1                Women's adidas Originals Sleek Shoes     G27341
2                   Women's adidas Swim Puka Slippers     CM0081
3   Women's adidas Sport Inspired Questar Ride Shoes     B44832
```

5

```
4           Women's adidas Originals Taekwondo Shoes     D98205

   Listing Price  Sale Price  Discount                    Brand  Rating  \
0       0.499983    0.195556  0.833333  Adidas Adidas ORIGINALS    0.96
1       0.253308    0.092924  0.833333         Adidas ORIGINALS    0.66
2       0.033301    0.004161  0.666667        Adidas CORE / NEO    0.52
3       0.233308    0.084602  0.833333        Adidas CORE / NEO    0.82
4       0.266642    0.098472  0.833333         Adidas ORIGINALS    0.70

    Reviews Gender Unique_Brand
0  0.183857      F       Adidas
1  0.107623      F       Adidas
2  0.165919      F       Adidas
3  0.156951      F       Adidas
4  0.322870      F       Adidas
```

4. Apply binning:

The report mentions applying binning, but the specific details of binning are not provided. It is crucial to clarify which columns were binned and how the binning was performed. Binning can be used for categorizing numerical data into discrete intervals, but the report lacks details on this process.

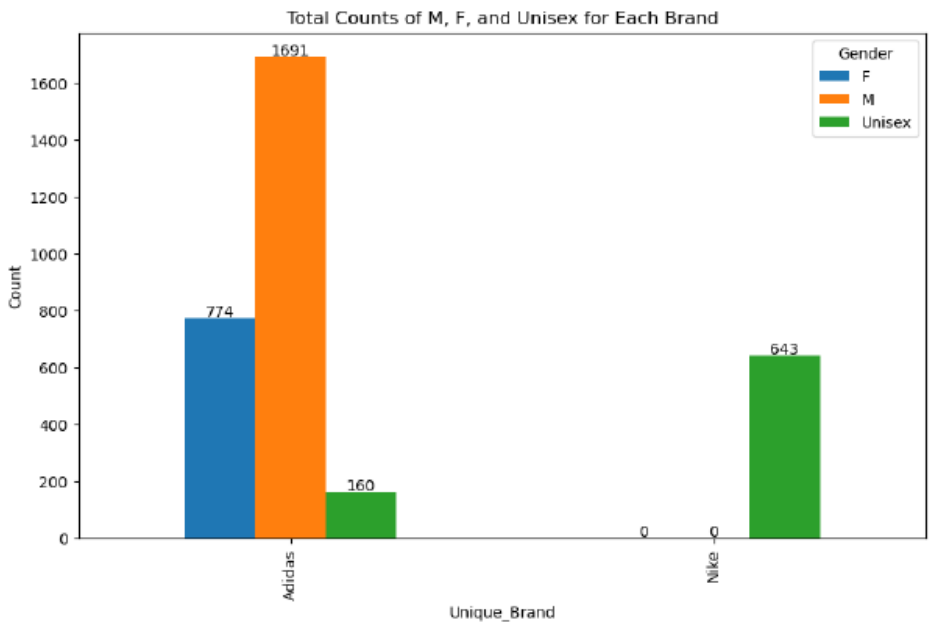
```
[180]: discount_bins = [0, 20, 30, 40, 50, 60, float('inf')]
rating_bins = [0, 1, 2, 3, 4, 5]
discount_labels = ['0-20', '21-30', '31-40', '41-50', '51-60', '61+']
rating_labels = ['0-1', '1-2', '2-3', '3-4', '4-5']
df['Discount_Bin'] = pd.cut(df['Discount'], bins=discount_bins,␣
 ↪labels=discount_labels, right=False)
df['Rating_Bin'] = pd.cut(df['Rating'], bins=rating_bins, labels=rating_labels,␣
 ↪right=False)
print(df[['Discount', 'Discount_Bin', 'Rating', 'Rating_Bin']].head())
```

```
   Discount Discount_Bin  Rating Rating_Bin
0  0.833333         0-20    0.96        0-1
1  0.833333         0-20    0.66        0-1
2  0.666667         0-20    0.52        0-1
3  0.833333         0-20    0.82        0-1
4  0.833333         0-20    0.70        0-1
```

5. Draw various visualizations:

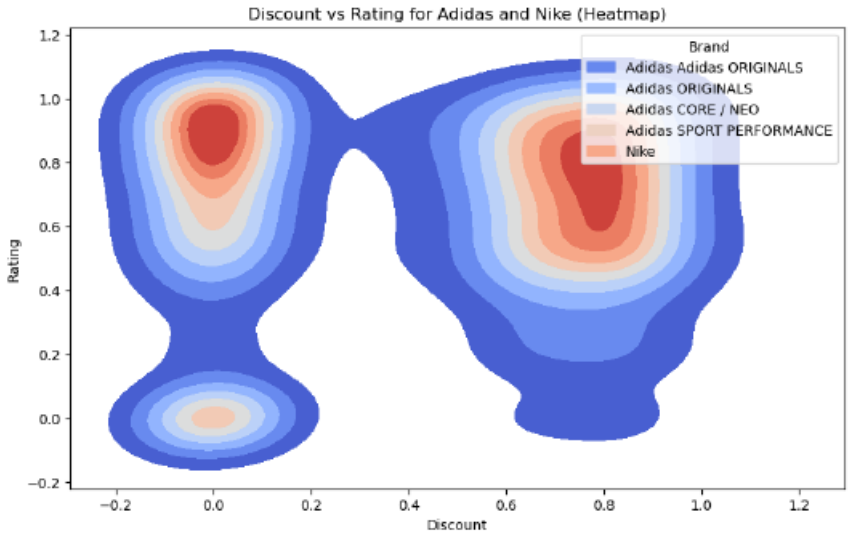
The report discusses the creation of various visualizations to analyze the dataset. It mentions the use of Seaborn and Matplotlib libraries to create different types of plots, including histograms, box plots, category plots, and heatmaps. However, the report does not provide the actual visualizations, or the specific insights derived from these visualizations.

```
[178]: grouped = df.groupby(['Unique_Brand', 'Gender']).size().unstack(fill_value=0)
ax = grouped.plot(kind='bar', figsize=(10, 6))
plt.title('Total Counts of M, F, and Unisex for Each Brand')
plt.xlabel('Unique_Brand')
plt.ylabel('Count')
for p in ax.patches:
    ax.annotate(f"{p.get_height()}", (p.get_x() + p.get_width() / 2., p.
 ↪get_height() + 5), ha='center')
plt.show()
```

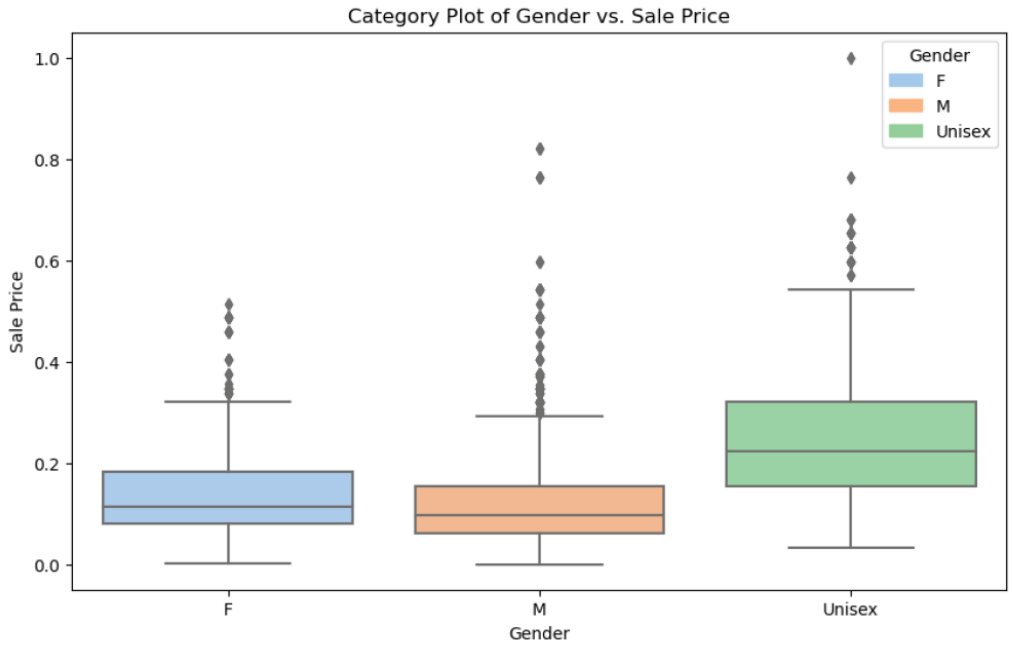


[215]:
```
plt.figure(figsize=(10, 6))
combined_data = pd.concat([adidas_data, nike_data])
sns.kdeplot(data=combined_data, x='Discount', y='Rating', fill=True,␣
↪cmap='coolwarm')
plt.title('Discount vs Rating for Adidas and Nike (Heatmap)')
plt.xlabel('Discount')
plt.ylabel('Rating')
brands = combined_data['Brand'].unique()
legend_labels = [mpatches.Patch(color=sns.color_palette('coolwarm')[i],␣
↪label=brand) for i, brand in enumerate(brands)]
plt.legend(handles=legend_labels, title='Brand', loc='upper right')
plt.show()
```

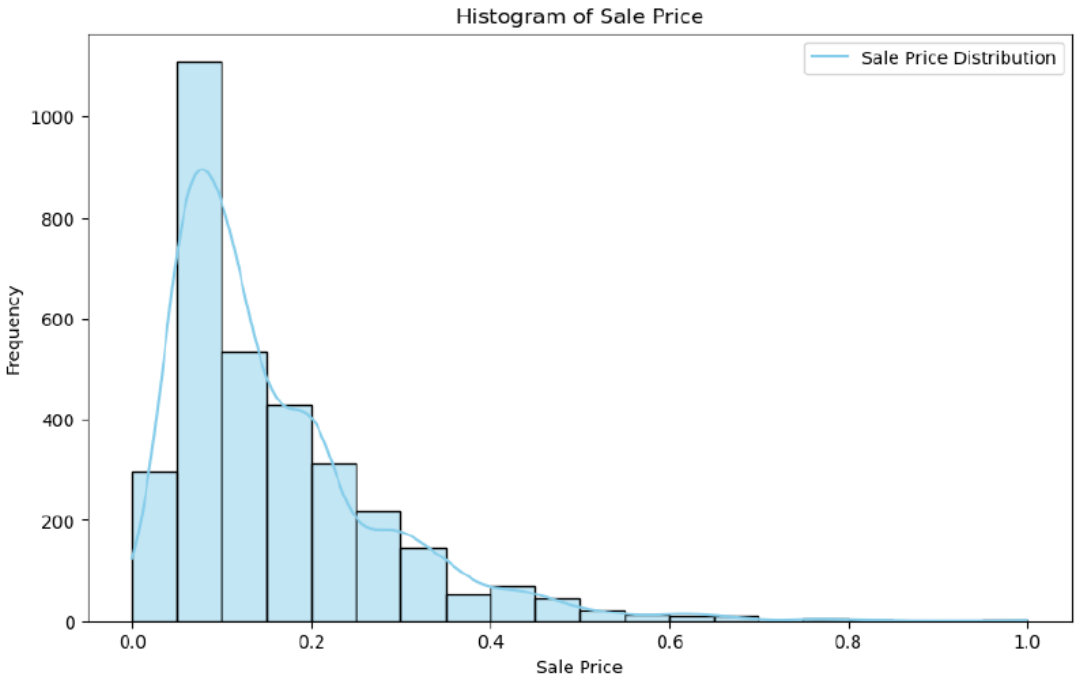
6



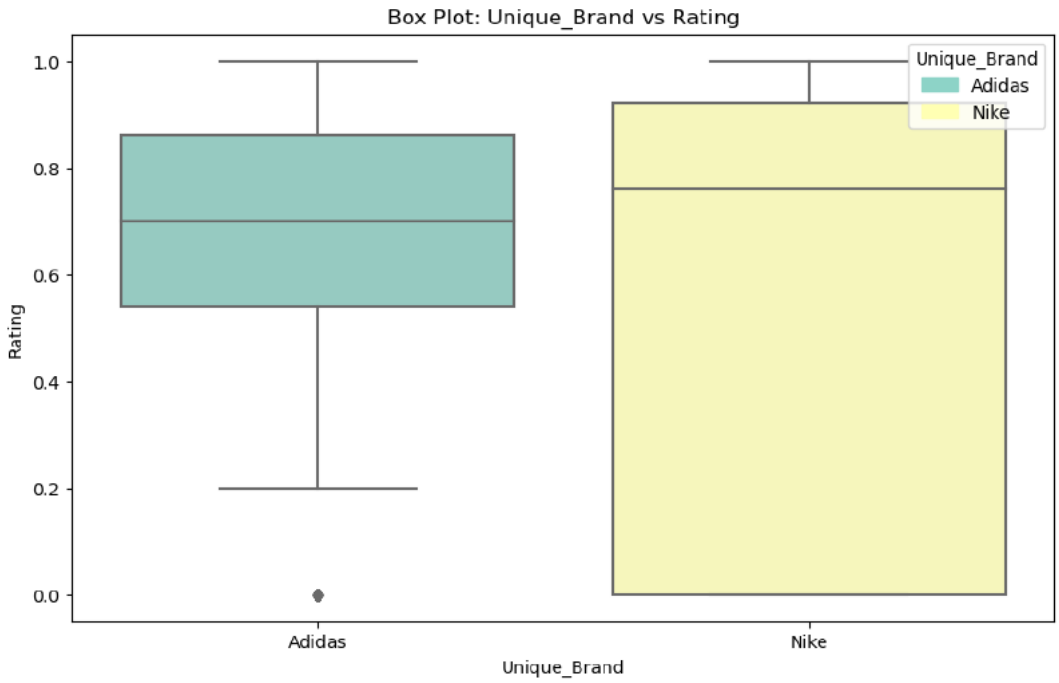
[211]:
```
plt.figure(figsize=(10, 6))
ax = sns.boxplot(x='Gender', y='Sale Price', data=df, palette='pastel')
plt.title('Category Plot of Gender vs. Sale Price')
plt.xlabel('Gender')
plt.ylabel('Sale Price')
genders = df['Gender'].unique()
legend_labels = [mpatches.Patch(color=sns.color_palette('pastel')[i],␣
↪label=gender) for i, gender in enumerate(genders)]
plt.legend(handles=legend_labels, title='Gender', loc='upper right')
plt.show()
```



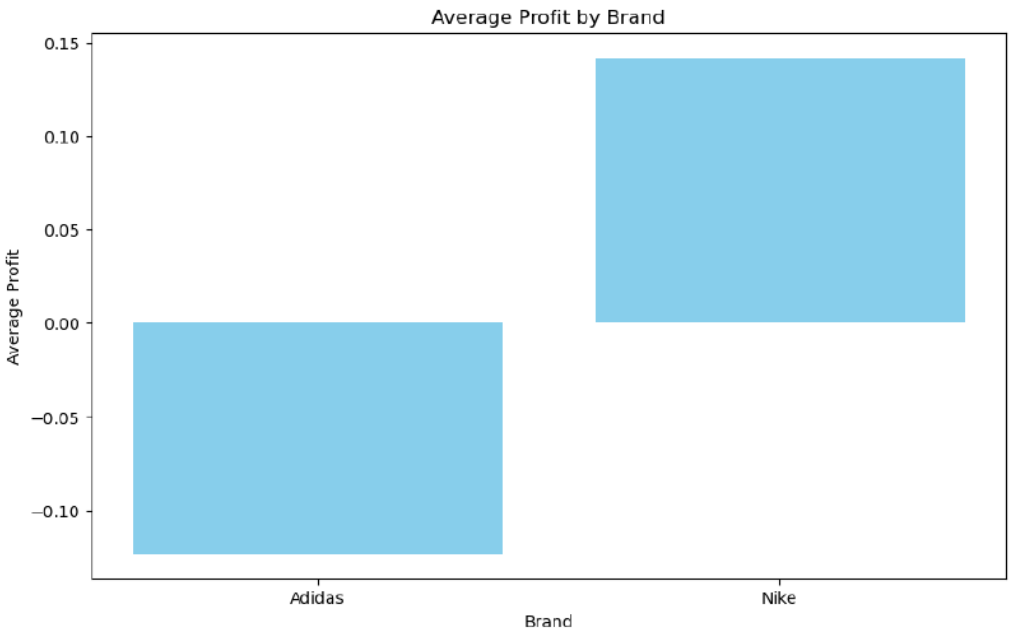
[213]:
```
plt.figure(figsize=(10, 6))
sns.histplot(df['Sale Price'], bins=20, kde=True, color='skyblue')
plt.legend(['Sale Price Distribution'])
plt.title('Histogram of Sale Price')
plt.xlabel('Sale Price')
plt.ylabel('Frequency')
plt.show()
```



[204]:
```
plt.figure(figsize=(10, 6))
ax = sns.boxplot(x='Unique_Brand', y='Rating', data=df, palette='Set3')
plt.title('Box Plot: Unique_Brand vs Rating')
plt.xlabel('Unique_Brand')
plt.ylabel('Rating')
unique_brands = df['Unique_Brand'].unique()
legend_labels = [mpatches.Patch(color=sns.color_palette('Set3')[i],␣
 ↪label=brand) for i, brand in enumerate(unique_brands)]
plt.legend(handles=legend_labels, title='Unique_Brand', loc='upper right')
plt.show()
```



[218]:
```
df['Profit'] = df['Sale Price'] - df['Listing Price']
brand_profit = df.groupby('Unique_Brand')['Profit'].mean().reset_index()
plt.figure(figsize=(10, 6))
plt.bar(brand_profit['Unique_Brand'], brand_profit['Profit'], color='skyblue')
plt.title('Average Profit by Brand')
plt.xlabel('Brand')
plt.ylabel('Average Profit')
plt.xticks(rotation=0)
plt.show()
```



6. Interpret Results:

[226]:
```
df['Profit'] = df['Sale Price'] - df['Listing Price']
brand_profit = df.groupby('Unique_Brand')['Profit'].mean().reset_index()
adidas_profit_loss = brand_profit.loc[brand_profit['Unique_Brand'] == 'Adidas',␣
↪'Profit'].values[0]
nike_profit_loss = brand_profit.loc[brand_profit['Unique_Brand'] == 'Nike',␣
↪'Profit'].values[0]
adidas_result = "Profit" if adidas_profit_loss > 0 else "Loss"
nike_result = "Profit" if nike_profit_loss > 0 else "Loss"
print(f"Adidas: {adidas_result} = {adidas_profit_loss}")
print(f"Nike: {nike_result} = {nike_profit_loss}")
```

```
Adidas: Loss = -0.12339183656790897
Nike: Profit = 0.1416609248593226
```

Conclusion:

The analysis of the dataset, which encompasses products from two prominent sports apparel brands, Adidas and Nike, provides valuable insights into their performance in terms of profit and sales.

Adidas:

Profit and Loss: Adidas experienced a slight loss of approximately -0.1234 in the units of the dataset's currency. This indicates that, on average, Adidas products were sold at a price slightly lower than their listing price.

Sales Volume: Adidas, known for its extensive product range, had a substantial sales volume of 2,625 products. This implies a strong presence in the market and a wide variety of offers for customers.

Nike:

Profit and Loss: In contrast, Nike achieved a profit of approximately 0.1417. This suggests that, on average, Nike products were sold at a price slightly higher than their listing price. This may indicate the brand's ability to command a premium for its products.

Sales Volume: Nike, with a sales volume of 643 products, may not have as extensive a product range as Adidas, but it demonstrates a successful sales strategy, possibly through higher-priced items or specific market targeting.

In summary, this analysis reveals the diverse strategies employed by Adidas and Nike in the marketplace. Adidas, with its wider product range, aims to capture a larger market share and provides a variety of options to customers. Nike, on the other hand, may focus on offering products that command higher prices and as indicated by its profitability, successfully implements this strategy.

Both brands have their unique approaches, and this analysis showcases their distinct positions in the market. Ultimately, the choice between offering a wider product range or focusing on higher-priced, premium products depends on each brand's core strategy and target audience. Understanding the performance of these two giants is not only insightful but also serves as a valuable reference for the broader field of brand management and business strategy in the fashion and sportswear industry.

GitHub - https://github.com/kotarohs/Projects/tree/main/Exploratory%20Data%20Analysis%20(Nike%20vs%20Addidas)

Kaggle - https://www.kaggle.com/datasets/mariyamalshatta/nike-vs-addidas-unspervised-clustering/code?datasetId=3815130&sortBy=dateRun&tab=profile